

Comprehensive Cancer Center of Metropolitan Health

Department of Oncology – Breast Multidisciplinary Tumor Board

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MULTIDISCIPLINARY BREAST TUMOR BOARD SUMMARY

Patient Name: Jenny M
Date of Birth: January 1, 1965 (Age 55 years)
Medical Record Number: MRN-8847521
Tumor Board Date: March 28, 2020
Presenter: Dr. Rachel Pemberton, Medical Oncology

Clinical Summary

Ms. Jenny M is a 55-year-old postmenopausal female with newly diagnosed invasive ductal carcinoma of the left breast. The patient initially presented with a palpable mass in the upper outer quadrant of the left breast detected on self-examination. Diagnostic mammography performed on March 2, 2020, revealed a 2.1 cm irregular, spiculated density in the left breast at 10 o'clock position, 6 cm from the nipple. Ultrasound-guided core needle biopsy was performed on March 5, 2020.

Pathology from Core Biopsy (March 10, 2020):

Diagnosis:

Invasive ductal carcinoma, Grade 2 (Nottingham score 6: tubule formation 3, nuclear pleomorphism 2, mitotic rate 1)

Biomarkers:

Estrogen Receptor (ER): Positive (90% of tumor nuclei, strong intensity)

Progesterone Receptor (PR): Positive (70% of tumor nuclei, moderate intensity)

HER2/neu: Negative by immunohistochemistry (0)

Ki-67 proliferation index: 18%

The patient has no significant family history of breast or ovarian cancer. She is otherwise healthy with no relevant comorbidities. Performance status is ECOG 0. Physical examination reveals a firm, mobile 2 cm mass in the left breast upper outer quadrant with no palpable axillary lymphadenopathy.

Imaging Review

Presented by Dr. Michael Torres, Diagnostic Radiology

Bilateral diagnostic mammography demonstrates a 2.1 cm irregular mass with spiculated margins in the left breast at the 10 o'clock position. No additional suspicious lesions identified in either breast. No pathologic calcifications. Targeted ultrasound confirms a 2.2 x 1.8 x 1.9 cm hypoechoic mass with irregular margins, posterior acoustic shadowing, and increased internal vascularity on Doppler imaging. The mass is located 6 cm from the nipple-areolar complex.

Axillary ultrasound examination shows morphologically normal-appearing lymph nodes bilaterally with preserved fatty hila and no cortical thickening. The largest visualized left axillary node measures 1.2 cm in longest dimension with normal architecture.

Breast MRI has not yet been performed. Discussion raised regarding the utility of MRI for extent-of-disease assessment and occult multifocality evaluation prior to surgical planning.

Pathology Review

Presented by Dr. Susan Choi, Anatomic Pathology

Pathology slides from the ultrasound-guided core needle biopsy were reviewed. The specimen shows infiltrating ductal carcinoma with moderate tubule formation, mild to moderate nuclear atypia, and low mitotic activity. The tumor cells are arranged in irregular nests and cords infiltrating into adipose and fibrous stroma. No lymphovascular invasion is identified in the sampled tissue. No ductal carcinoma in situ (DCIS) component is appreciated in the cores obtained.

Immunohistochemical stains confirm strong, diffuse ER positivity (Allred score 7), moderate PR positivity (Allred score 6), and HER2 negativity (0 by IHC, no membrane staining). Ki-67 labeling is positive in approximately 18% of tumor nuclei, consistent with intermediate proliferative activity.

The biomarker profile indicates a luminal A-like breast cancer. Oncotype DX recurrence score testing on the surgical specimen would provide additional prognostic and predictive information to guide adjuvant systemic therapy decisions.

Surgical Oncology Review

Presented by Dr. Jennifer Caldwell, Surgical Oncology

From a surgical standpoint, the patient is an excellent candidate for breast-conserving therapy. The tumor-to-breast size ratio is favorable, and the lesion is well-localized in the upper outer quadrant. The patient expresses strong preference for breast preservation if oncologically appropriate.

The surgical plan would include lumpectomy with standard margins (no tumor on ink) and sentinel lymph node biopsy. Given the clinically and radiographically negative axilla, sentinel node biopsy is appropriate for staging. If sentinel nodes are negative, no further axillary surgery would be required. If macrometastatic disease is identified in sentinel nodes, completion axillary lymph node dissection versus axillary radiation would be discussed based on tumor burden and patient factors.

Preoperative MRI staging remains under consideration. While not mandatory, it may identify occult disease that could alter surgical approach in approximately 15-20% of cases. However, false-positive findings can also lead to unnecessary additional biopsies.

Medical Oncology Review

Presented by Dr. Rachel Pemberton, Medical Oncology

This is a stage I-II, ER-positive, HER2-negative breast cancer. The neoadjuvant versus adjuvant approach was discussed. Given the tumor size (just over 2 cm), favorable biology (grade 2, hormone receptor positive, HER2 negative), and clinically negative nodes, upfront surgery is the preferred approach. Neoadjuvant chemotherapy is typically reserved for larger tumors, locally advanced disease, or cases where tumor downsizing is needed for breast preservation—none of which apply in this case.

Following surgery, adjuvant systemic therapy recommendations will be guided by final pathologic staging and Oncotype DX recurrence score. If the tumor is node-negative or has only micrometastatic disease and the Oncotype score is low (<18), adjuvant endocrine therapy alone would be appropriate. Intermediate scores (18-30) require shared decision-making regarding chemotherapy benefit. High scores (>30) or node-positive disease would favor adjuvant chemotherapy followed by endocrine therapy.

Standard endocrine therapy for a postmenopausal woman would be an aromatase inhibitor (such as anastrozole, letrozole, or exemestane) for 5-10 years.

Radiation Oncology Review

Presented by Dr. Thomas Wei, Radiation Oncology

Following breast-conserving surgery, whole-breast radiation therapy is standard of care to reduce local recurrence risk. The typical regimen is 45-50 Gy to the whole breast in 1.8-2.0 Gy fractions with a boost of 10-16 Gy to the tumor bed. Hypofractionated regimens (42.5 Gy in 16 fractions or 40 Gy in 15 fractions) are also standard options with equivalent oncologic outcomes and acceptable cosmesis.

Regional nodal irradiation would be considered if ≥ 4 positive nodes are identified, or if 1-3 positive nodes are present with additional high-risk features. Partial breast irradiation is not recommended in this case given the patient's age under 60 and grade 2 histology.

Multidisciplinary Discussion

The case was discussed at length with participation from all disciplines. Key discussion points included:

- **Surgical Approach:** Consensus strongly favored upfront surgery with breast-conserving therapy. The patient is an excellent candidate for lumpectomy with sentinel lymph node biopsy. Neoadjuvant chemotherapy would not provide significant benefit given favorable tumor biology and size.
- **Role of Preoperative MRI:** The benefits and limitations of breast MRI were reviewed. While MRI can detect occult multifocal/multicentric disease, it also has a false-positive rate of 10-25% and may delay surgery. The consensus was to offer MRI as an optional study but not mandate it. The patient's young age (55) and dense breast tissue slightly favor obtaining MRI, but the decision should be individualized based on patient preference and timeline considerations.
- **Sentinel Node Mapping:** Given clinically and radiographically negative axilla, sentinel lymph node biopsy is the appropriate staging procedure. Dual tracer technique (radioisotope and blue dye) recommended for optimal node identification.

- **Adjuvant Therapy Planning:** Final adjuvant systemic therapy recommendations should await surgical pathology and Oncotype DX results. The tumor's favorable biomarker profile (ER/PR positive, HER2 negative, grade 2) suggests endocrine therapy will be the cornerstone of treatment, with chemotherapy reserved for higher-risk features.
- **Genetic Testing:** Given the patient's age at diagnosis (55) and lack of significant family history, routine BRCA testing is not indicated. However, if family history details emerge suggesting hereditary risk, genetics referral can be arranged.

Consensus Recommendations

1. **Surgery:** Proceed with left breast lumpectomy and sentinel lymph node biopsy. Target surgical margins: no tumor on ink. Consider preoperative breast MRI for extent-of-disease assessment (optional, patient preference-driven).
2. **Pathology:** Submit surgical specimen for Oncotype DX recurrence score testing (if node-negative or micrometastatic disease only) to guide adjuvant chemotherapy decision-making.
3. **Adjuvant Systemic Therapy:** Defer final recommendations pending surgical pathology and genomic testing results. Anticipated approach:
 - If node-negative and Oncotype DX low risk: adjuvant endocrine therapy with aromatase inhibitor for 5-10 years
 - If intermediate risk: shared decision-making regarding chemotherapy benefit
 - If high risk or node-positive: adjuvant chemotherapy (regimen such as dose-dense AC-T or TC) followed by endocrine therapy
4. **Radiation Therapy:** Whole-breast radiation therapy to follow surgical recovery and any indicated chemotherapy. Standard fractionation (50 Gy in 25 fractions) or hypofractionated regimen (42.5 Gy in 16 fractions) with tumor bed boost.
5. **Follow-up:** Re-present to tumor board following surgical pathology if any unexpected findings or clinical questions arise. Patient will be scheduled for routine multidisciplinary survivorship care following completion of primary treatment.

Attendees

Dr. Rachel Pemberton, MD – Medical Oncology (Presenter)
 Dr. Jennifer Caldwell, MD, FACS – Surgical Oncology
 Dr. Susan Choi, MD – Anatomic Pathology
 Dr. Michael Torres, MD – Diagnostic Radiology
 Dr. Thomas Wei, MD – Radiation Oncology
 Dr. Amanda Foster, MD – Breast Imaging
 Ms. Patricia Nguyen, RN, MSN – Nurse Navigator, Breast Program
 Ms. Sarah Klein, PharmD, BCOP – Oncology Pharmacy

Documented by:

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Medical Oncology

Comprehensive Cancer Center of Metropolitan Health

Date: March 28, 2020

This summary represents the multidisciplinary consensus recommendations discussed at the Breast Tumor Board. Individual patient care decisions should be made in consultation with the patient's treating physicians and may vary based on evolving clinical circumstances, patient preferences, and additional diagnostic information.